CSE2525 · DATA MINING · 2026–27

# Distances & Similarity

*Choosing a distance is choosing which patterns an algorithm can see — worked through two case studies.*

A. Anand · study deck

PART I

# Why distance underlies data mining

*Before any formula: what does a distance actually let us do?*

THE ONE ARGUMENT OF THIS LECTURE

# Mining is deciding what belongs together

CENTRAL IDEA

Data mining discovers patterns by deciding which objects belong together. Representation and distance are what make that decision precise.

So the interesting question isn't "what's the formula" — it's *which differences we choose to let matter*.

Provocation to hold onto: can we change the discovered patterns without changing the data?

DISTANCE IS A SHARED BUILDING BLOCK

# One dataset — four questions




| Mining task | The question, expressed through distance |
|---|---|
| Classification | Which labelled examples are closest to this new object? |
| Clustering | Which objects are close enough to form groups? |
| Anomaly detection | Which objects are far from the usual observations? |
| Similarity search | Which existing objects are closest to this query? |

DON’T OVERCLAIM

# Distance is a common language — not a law

HONEST QUALIFICATION

Many methods use distance **explicitly**. Others use related notions — **similarity**, **reconstruction error**, **divergence**. Distance gives a shared vocabulary, but *not every method requires a metric*.

We’ll still see that changing the distance changes the neighbours, the groups, and what counts as far.

PART II

# Task → representation → distance

*The choice of distance is the model — so make the choice deliberately.*

“SIMILAR” IS AMBIGUOUS ON PURPOSE

# Close in what sense?

— Two shoes can be close in price but different in purpose.

— Two products can have similar descriptions but very different prices.

— Two visually similar products may be substitutes; two bought-together products may be complements.

CONSEQUENCE

Different features give different neighbours. Picking the feature and the distance *is* picking the model — this is the most important idea in the lecture, so we put it first.

FOUR DECISIONS, IN ORDER

# Task → representation → distance → decision

| Choice | What it determines |
|---|---|
| Task | What “similar” should mean |
| Representation | What information is available for comparison |
| Distance | Which differences matter, and how much |
| Decision rule | What we do with the resulting distances (e.g. top-k) |

And you don’t always need a vector first — trees, strings and sets have distances directly.

PART III

# Metrics as useful guarantees

*What properties make a distance predictable — and fast?*

EACH AXIOM BUYS YOU SOMETHING

# The axioms, with their payoff

| Property | What it guarantees |
| --- | --- |
| Non-negativity $d(x,y) \geq 0$ | Difference has a meaningful zero baseline |
| Identity of indiscernibles $d(x,y)=0 \Leftrightarrow x=y$ | Distinct objects stay distinguishable |
| Symmetry $d(x,y)=d(y,x)$ | Comparing A to B equals B to A |
| Triangle inequality $d(x,z) \leq d(x,y)+d(y,z)$ | Known distances constrain unknown ones |

WHY THE TRIANGLE INEQUALITY EARNS ITS KEEP

# Metric structure lets us skip work

PRUNING WITH THE TRIANGLE INEQUALITY

A query $q$ is distance 10 from a reference $r$. A candidate $c$ is distance 2 from $r$. Then $c$ must be at least **10 − 2 = 8** from $q$.

SO

If we only want neighbours within distance 3, we can discard $c$ without ever computing d(q,c).

That is the computational reason metric structure matters — it powers indexes that avoid the full $n^2$ comparisons.

KEEP THESE SEPARATE

# Valid metric ≠ useful distance

ASK BOTH, INDEPENDENTLY

1 · Is it a mathematically valid metric?  2 · Does it capture the similarity our task needs?

Neither answer settles the other. A valid metric can be useless for a task; a genuinely useful dissimilarity need not satisfy every axiom. We'll see both in the case studies.

PART IV · CASE STUDY 1

# Products: which differences matter

*Designing a similarity, as a sequence of decisions.*

"I WANT TO SUGGEST SIMILAR ITEMS"

# First decide what relevance means




What makes a product a useful recommendation for another — shared category? co-purchase? "related items" / clicks vs. purchases define the target before any distance is chosen.

WHAT INFORMATION IS AVAILABLE

# Four representations of a product

### Category

- Where it sits in the catalogue tree

### Text

- Title & description terms

### Price

- A single numeric feature

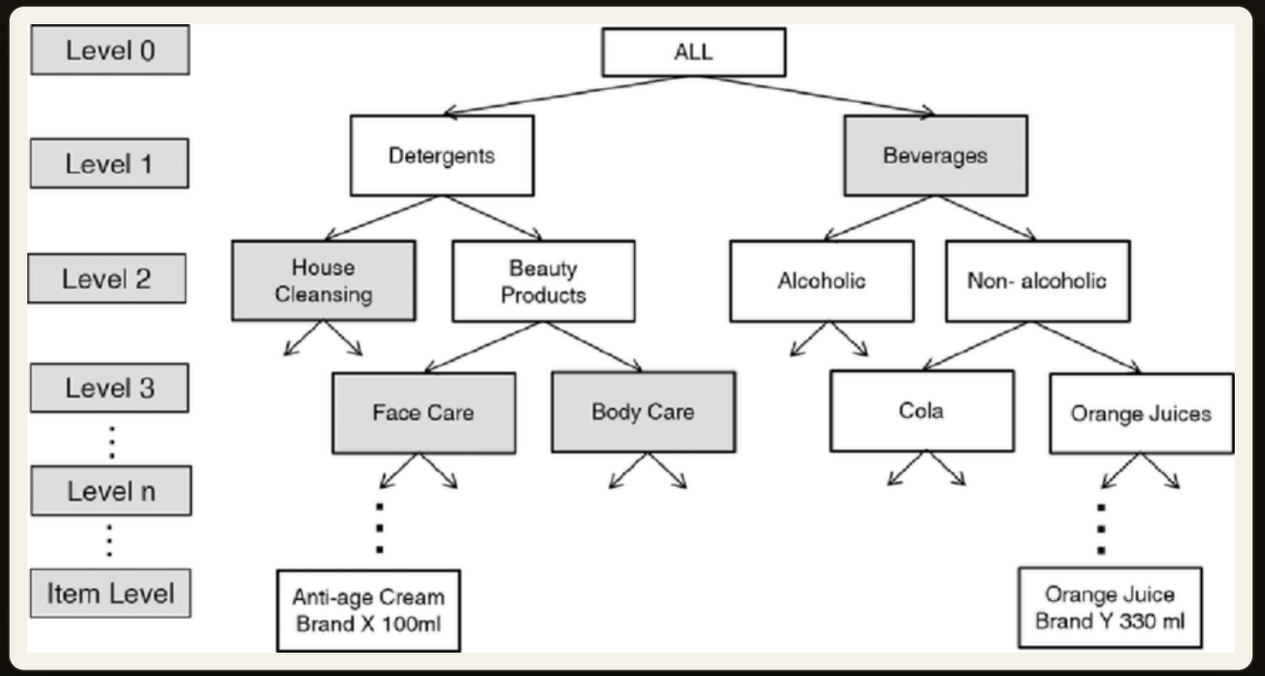
### Reviews

- Ratings & their distribution

Each representation needs its own distance — and each will disagree about who the neighbours are.

CATEGORY → A TREE

# Taxonomy distance = shortest path




Distance = path length up to a common ancestor and back down. A metric on category nodes. But distinct products in the same category have distance 0 — the representation, not the metric, is the limit.

TEXT → DIRECTION OF A TERM VECTOR

# Cosine: similar wording, any length

$$\cos(\theta) = (a \cdot b) / (\|a\| \, \|b\|) \quad d_{cos} = 1 - \cos(\theta)$$

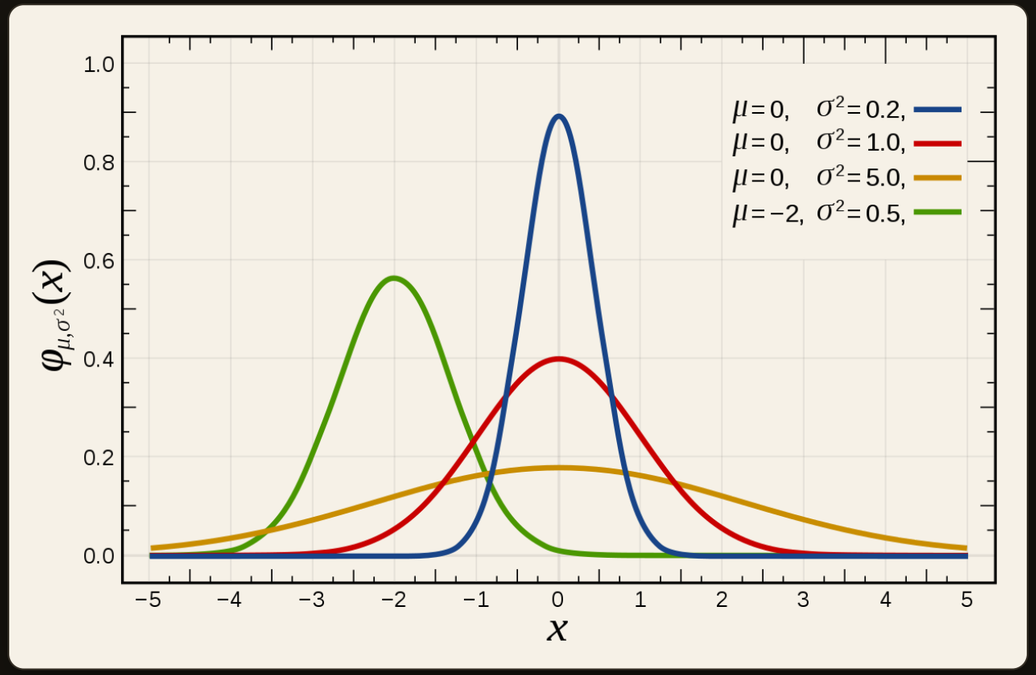
READ IT CAREFULLY

Only the direction of the term vectors matters, never their length — great for descriptions of different sizes.

Mind two traps: distance 0 means same direction, not identical (proportional vectors collapse to 0); and $1 - \cos\theta$ is not a true metric — it can violate the triangle inequality. Useful, but not a metric.

PRICE → EUCLIDEAN, BUT BEWARE UNITS

# Standardize — then don’t over-trust it




Raw price dominates small-scale features, so standardize: z = (x − μ)/σ. But equal scale ≠ equal relevance, and it does not remove outlier sensitivity — scaling is not a priority decision.

REVIEWS → SPARSE & COLD-START

# What about products with no history?




Most products have few reviews; brand-new ones have none (cold start). The distance must degrade gracefully when a representation is missing or thin.

QUESTION

Your query is a **running shoe**. Ranked by **price**, the single closest product is a formal **dress shoe**. Is that a good recommendation — and what decides the answer?

*Same three candidates, three different distances.*

Think — then → for the answer.

SEE THE CHOICE CHANGE THE ANSWER

# One query, three candidates, three rankings

QUERY

a **running shoe** · €120 · *Footwear › Running* · “lightweight breathable running shoe”.

| Candidate | By price | By category | By text |
|---|---|---|---|
| A · trail running shoe (€130) | near (Δ10) | nearest — same “Running” | near — “running shoe” |
| B · running socks (€15) | far (Δ105) | far — Apparel › Socks | near — “breathable lightweight running” |
| C · formal dress shoe (€125) | nearest (Δ5) | mid — shares “Footwear” | far — only “shoe” |

Price crowns a dress shoe; text pulls in socks; only the right representation ranks the other running shoe first. The distance is the model.

FROM OBJECTIVE TO ALGORITHM

# Combine distances — weights are priorities

$$d(x,y) = w_c \cdot d_{category} + w_t \cdot d_{text} + w_p \cdot d_{price}$$

TWO WARNINGS THAT RIDE TOGETHER

The terms must be on comparable scales before you add them — exactly the standardization point again.

The weights w encode priorities; they are the model's hyperparameters. Standardization does not choose them for you — you decide what should matter. Then rank the top-k neighbours and evaluate against real clicks/purchases.

PART V · CASE STUDY 2

# Videos: which differences to ignore

*Invariance, thresholds, and a system that scales.*

YOUTUBE: NEAR-DUPLICATES OF AN UPLOAD

# Equivalent for our task, different in the bytes



Figure 1. Examples of queries and near-duplicate videos from CC_WEB_VIDEO dataset.

What should be ignored: re-encoding, modest colour changes, logos/overlays, small crops. The job is the opposite of case 1 — choosing differences to discard.

BYTE EQUALITY IS THE WRONG TOOL

# Exact ≠ near

THE FAILURE

Byte equality and **cryptographic hashes** (MD5, SHA) are designed so one changed pixel flips the whole digest. They detect **identical files**, never **perceptually similar** ones.

We need a representation where small perceptual changes → small distance.

CONSTRUCT THE RIGHT REPRESENTATION

# Perceptual hashing (pHash)

THE GENERAL IDEA

Reduce each frame to a short fingerprint that survives re-encoding and minor edits — typically from low-frequency structure, so cosmetic changes don’t move it.

Keep the idea separate from any one scheme. Code length, keyed behaviour and non-invertibility are properties of *particular* hashes, not defining features of perceptual hashing in general.

COMPARE → HAMMING DISTANCE

# Count the differing bits

$a = 10110010 \quad b = 10111010 \rightarrow d_{Hamming} = 1$

BACK TO PART III

Hamming distance is a proper metric on fingerprints — all four axioms hold.

Yet distinct videos can share a fingerprint. A metric *in the representation space* does not guarantee the originals are distinguishable — the same lesson as taxonomy-distance on products.

MAKE A DECISION, ACCEPT ERRORS

# Pick a threshold — trade off the two mistakes

DECISION RULE

call it a near-duplicate when Hamming distance ≤ **τ**.

- Threshold too tight → miss real duplicates (false negatives).
- Threshold too loose → flag unrelated videos (false positives).
- τ is chosen from data — exactly the kind of knob the task, not the metric, decides.

MAKE IT SCALE

# Fast filter → slow check

*cheap candidate generation → expensive exact verification*

THE RECIPE (YOU'LL SEE IT AGAIN)

Generate a small candidate set cheaply (fingerprint buckets / LSH), then verify the few survivors with the expensive comparison. Never materialise all $n^2$ distances.

The same shape powers web search, retrieval and question answering — a recurring pattern this year.

SYNTHESIS

# Change the distance, change the patterns

*Why this one choice ripples through every task.*

BACK TO THE FOUR TASKS

# Changing the distance changes the answer

- Nearest neighbours change → classifications and recommendations change.
- Neighbourhoods change → clusters and density estimates change.
- What counts as “far” changes → detected anomalies change.

| Case | Main lesson |
| --- | --- |
| Products | Choosing which differences matter |
| Videos | Choosing which differences to ignore |

TAKE AWAY · AND A TRANSFER EXERCISE

# Choosing a distance is choosing what you can see

THE MESSAGE

Task → representation → distance → decision. Metrics buy predictability and speed; but a valid metric and a task-useful distance are two separate questions.

Choosing a distance is choosing which patterns an algorithm can see.

Transfer: for detecting duplicate news articles — what representation would you choose, which changes should be ignored, and how would you evaluate the result?